

Title of Report

Hotel/Hospitality Management System

Contents

1.Abstract.....	4
2.Introduction.....	4
2.1 Project Overview.....	4
2.2 Aims & Objectives.....	4
2.3 A Brief overview of the hospitality and hotel sector.....	5
2.3.1 Food and Beverage Industry.....	5
2.3.2 Lodging Industry.....	5
2.3.3 Recreation Industry.....	6
2.3.4 Travel and Tourism Industry.....	6
2.3.5 Meeting and Events Industry.....	6
2.4 Project Deliverables.....	6
3. Background review/Literature Review.....	7
3.1 Historical Background.....	7
3.2 Design Process for Website Creation.....	7
3.2.1 Information Gathering on hotel websites.....	8
3.2.2 Planning.....	10
3.2.3 Design.....	10
3.2.4 Content.....	12
3.2.5 Functionality.....	13
4. Methods & Development Approach.....	14
4.1 The Waterfall Methodology.....	14
4.2 Languages Involved.....	15
4.3 Software and tools for website creation.....	16
4.4 SQL / Database Platform.....	18
4.4.1 Define the objective of your database.....	18
4.4.2 Locate and consolidate the necessary data.....	18
4.4.3 Distribute the data into tables.....	18
4.4.4 Change data items into columns.....	18
4.4.5 Identify primary keys.....	19
4.5 Design approach for hotel website.....	19
4.6 Wire Frames Developed.....	20
4.7 Prototypes Developed.....	22

5. Implementation.....	23
5.1 Visual Studio code setup.....	23
5.2 Database Initialising and Table creation.....	24
5.3 Developing Front-end and backend (PHP).....	26
5.3.1 Homepage.....	26
5.3.2 Rooms Page.....	26
5.3.3 Payment Page.....	27
5.3.4 Manage Booking Page (Guests).....	27
5.3.5 Login Page.....	28
5.3.6 Manage Booking Page (Employees).....	28
5.3.7 View Bookings Page.....	29
5.3.8 Connection file (conn.php)	29
5.4 Refinement and finalisation.....	29
6. Analysis/ Evaluation.....	30
6.1 Usability testing.....	30
6.1.1 Planning.....	30
6.1.2 Conducting.....	30
6.1.3 Analysing.....	31
6.1.4 Reporting.....	31
6.1.4.1 Functionality and Navigation.....	31
6.1.4.2 Visual Presentation.....	31
6.1.4.3 Recommendations.....	32
7. Conclusions & Recommendations.....	32
7.1 Effective Utilization of Tools and Technologies.....	32
7.2 Structured Database Design.....	32
7.3 User Centric Interface Design.....	32
7.4 Thorough Testing and Evaluation.....	32
7.5 Future Enhancement and Recommendations.....	33
8. List of references.....	33
9. Appendices.....	35
9.1 Screen shot of code.....	35
9.2 Screen shot of navigation bar.....	37

1. Abstract

The report details the development and implementation of a hotel management software project, emphasizing the utilization of tools such as PHP, MySQL, and phpMyAdmin, alongside user-centered design principles. Through structured database design and user-centric interface development, the project aimed to output hotel operations, enhance guest experiences, and optimize employee workflows. Usability testing, guided by an expert in the field of usability testing, provided insights for refinement, highlighting areas for improvement in functionality, navigation, and visual presentation. Recommendations for future enhancements include streamlining payment processes, improving user guidance, and continuing to optimize system performance based on user feedback. Overall, the project demonstrates effective utilization of technology to create a robust and user-friendly hotel management solution.

2. Introduction

In an era marked by technological advancements and evolving consumer preferences, the hospitality industry stands at the forefront of innovation. As we witness the paradigm shift towards digitalization, the significance of seamless hotel management systems cannot be overstated. This report presents an overview of the development process and key components of a comprehensive hotel management website.

Hotel management is the practice of overseeing every operation of the business to ensure consistent growth and development. This can involve the management of anything related to the hotel industry and requires knowledge of finance and accounts, customer service, staff management, marketing, hotel administration and more. (Elphick. D, 2024)

2.1. Project Overview

The topic I have chosen is hospitality or hotel management. The purpose of managing a hotel management website is to successfully establish a constant flow of travellers and guests to the property throughout the period, while also showcasing the wide variety of services and products.

The primary reason to choose the topic of hotel management due to my prior experience working in the hospitality industry. Having served as an employee managing bookings and guest services, I developed a deep understanding of the challenges and complexities involved in ensuring smooth hotel operations. This firsthand experience sparked my interest in exploring how technology could enhance these processes.

Working in the fast-paced environment of a hotel, I witnessed the complexities of managing reservations, coordinating with various departments, and ensuring guest satisfaction. I found myself interested by the potential for software solutions to revolutionize these operations, making them more efficient and guest centric.

2.2. Aims & Objectives

The aim of the hotel management system software is to provide a comprehensive and userfriendly platform for both customers and the management team to efficiently manage hotel

bookings and operations by reducing the burden of managing complex data and GUI on the website for both kind of users.

For customers and primarily for aged customers, the software aims to offer a seamless and convenient booking experience, allowing them to easily search for available rooms, view amenities and pricing, make reservations, and manage their bookings. Additionally, it provides features such as online payment processing, reservation modifications, and cancellation options, all accessible through an intuitive and responsive user interface.

For the management team, the software aims to streamline hotel operations by providing tools for managing room inventory, tracking reservations, and optimizing room allocations. It also offers functionalities for staff to view, and update booking details, manage room availability, and oversee various aspects of hotel management.

The objective is to meet the above aims by having comprehensive research the initiative was conducted, encompassing extensive exploration across diverse web platforms. This attempt involved thorough analysis of various online resources, spanning industry-leading websites, academic repositories, professional forums, and relevant literature sources. By delving into a range of digital platforms, gathering best practices, and extract invaluable knowledge pertinent to the development objectives at hand. Several computer languages and tools were employed This approach ensured a well-rounded understanding of the landscape, enabling us to leverage cutting-edge methodologies and incorporate innovative solutions into the hotel management system's design and functionality. Furthermore, the selection of coding and functional languages played a vital role, particularly as data management emerged as a priority following web design. This consideration came into play immediately after GUI formation, underscoring the significance of seamless integration between design and functionality for effective data handling and processing.

2.3. A Brief overview of the hospitality and hotel sector

The hospitality industry refers to various businesses and services linked to leisure and customer satisfaction. A defining aspect of the hospitality industry is that it focuses on ideas of luxury, pleasure, enjoyment, and experiences instead of catering to necessities and essentials. **(Martijn, 2019).**

The basic theme is to provide services to the guests. The hotel industry is defined as a business sector that evolves around providing accommodation to guests. The hotel industry consists of subdivisions such as lodging service, food & beverage service, tourism, transportation, and events. These subdivisions often overlap and collaborate to create an overall guest experience. As a core component, customer experience and service excellence remain the key focus in the hotel industry. Hotels in the industry can regain repeat stays by improving restaurant operations for better customer experience and customer service. **(F.H.A, 2024)**

There are various categories in which hotel sector operates to deliver hospitality for their guests some are mentioned below: -

2.3.1. Food and Beverage Industry

The food and beverage industry consists of the preparation, transportation, and serving of food or beverage to customers. Given the wide variety of options available to customers when it comes to food, this sector has become one of the most successful and easy to identify sectors of the industry.

2.3.2. Lodging Industry

The lodging industry consists of businesses that provide overnight accommodations to guests. Depending on the type of service you provide, businesses in the lodging industry can accommodate as few as one or two guests to as many as hundreds. Because of this large range, your experience across different segments of the lodging industry can vary immensely.

2.3.3. Recreation Industry

The recreation sector of the hospitality industry offers various entertainment, hobby, and educational opportunities for customers. Typically lasting just a few hours, these experiences include activities like day trips and leisure pursuits. Revenue is often generated through admission tickets, concessions, and food and beverage sales.

2.3.4. Travel and Tourism Industry

Travel and tourism are one of the largest sectors of the hospitality industry. In general, travel and tourism allow people to visit destinations outside of their normal environment. Distance travelled and length of stay vary significantly depending on the context. Some shorter trips can be as brief as a day, such as sleeping overnight while visiting friends or family.

2.3.5. Meetings and Events Industry

This segment of the hospitality industry consists of a variety of social, business, and sporting events. These events can range from something as intimate as a small wedding to events on a much larger scale, such as an expo. Venues that support the meetings and events industry have become integral to most major cities, not just because of the flexibility they offer, but because of the potential revenue they can bring in. (Kurtz, J, 2022)

2.4. Project Deliverables

In the project deliverables, the aim is to provide tangible outcomes that reflect the successful completion of the hotel management system software project. These deliverables encompass various aspects of the development process.

Primarily, this report will serve as a comprehensive guide outlining the functional and nonfunctional requirements of the software. This document will detail user stories, use cases, and system constraints, providing a roadmap for the development of the software.

Though, further it will be displayed how user interface (UI) design mock-ups, presenting highfidelity representations of the software's graphical user interface (GUI) seems to be. These mock-ups will showcase the layout, design, and user interactions of key screens, such as the homepage, room booking interface, and reservation management dashboard.

A detailed backend system architecture diagram will thoroughly be explained, illustrating the components, modules, and data flow of the backend system. This diagram will include the database schema, server architecture, and integration points with external systems, ensuring a clear understanding of the software's underlying infrastructure.

Furthermore, the software delivers a structured database schema and data dictionary, outlining the database tables, fields, relationships, and constraints necessary for efficient data storage and retrieval.

Finally, a functional software prototype will be developed, demonstrating core features and functionalities of the hotel management system.

3. Background review/Literature Review

3.1. Historical Background

In the beginning, there was a fire at the hearth, a warm meal and a roof over one's head. Thus begins the history of the hotel industry. People have been providing hospitality for weary travellers since our ancestors inhabited the earth about 100,000 to 200,000 years ago. Some years down the line, in the age of antiquity, spas and bathhouses often provided sleeping facilities and were popular attractions for those seeking rest and relaxation.

In medieval times, it was mostly cloisters and abbeys that provided travellers with accommodation. It was also during this period that more and more inns and guesthouses were opened on popular trade routes to provide food and lodging for travelling merchants and their horses. These were not hotels as we know them today. Hotel chains such as the "Ritz" only came into being hundreds of years later and were a manifestation of the growing prosperity of developing industrial nations.

The technological, economic, and social developments in the 19th and particularly the 20th century led to huge changes in the hotel industry. Travelling was no longer just for the privileged upper classes. The invention of trains and automobiles as well as social benefits such as holidays and reduced working hours made it possible for the average person to indulge in travel.

The second half of the 20th century saw a huge economic boom in industrial nations which led to increased travel amongst the larger populace. It wasn't just newly acquired wealth that promoted more travel the population explosion, urbanisation and demographic developments all contributed to an increased desire to travel and the growth of the tourism and hotel industry. Because of this increased demand, many more guesthouses, hostels, motels, or international hotel chains that catered to various target groups and in different price categories opened all over the world. (Customer Alliance, n.d.).

73 years ago, the first hotel reservation system was created by Westin (1947), Hotel type. Prior to this, hotels were known to be using a series of books and pages to keep track of the guests who were booking rooms and who were checking into the hotel. Given just how busy the hotel industry is today, it is almost impossible to imagine what it would be like to have to take care of everything with a register at the front desk. Now when bookings are made via many different points of sale, it's a struggle to believe that for years, even until 2010, reservations were made by telephone, telegram, or post. (Asksuite, 2021).

3.2. Design Processes for Website Creation

As the focus was to improvise UI/UX of the software by implementing robust website development techniques several procedures and standards were carried out. Moreover, the process was carried out into three different parts which was constructing front-end, styling it and overall functioning of the software.

The standard development process that I have followed: -

- Information Gathering
 - Planning
 - Design
 - Content
 - Functionality
- (www.Mailchimp.com, 2020)

3.2.1 Information Gathering on hotel websites.

The research commenced with thorough exploration and analysis of existing platforms offering similar services to users. Identifying the shortcomings within their software became paramount, as I aimed to rectify these errors and prevent their recurrence in my website.

The website I have researched are: -

- **About Travel Lodge**

Travelodge is a well-known hotel chain with a strong presence across the United Kingdom, Ireland, and Spain. They pride themselves on offering comfortable accommodations at affordable prices. With numerous hotels located in key cities and popular travel destinations, Travelodge aims to make traveling convenient and accessible for everyone. Their user-friendly website makes it easy for guests to book rooms and manage their reservations. Travelodge also provides essential amenities like free Wi-Fi and parking to enhance the guest experience.

- **Pros**

Travelodge's website is user-friendly, making it easy for guests to navigate and find the information they need. The booking process is straightforward, allowing guests to quickly reserve rooms and manage their reservations online. Additionally, the website provides detailed descriptions and images of each property, helping guests make informed decisions about their accommodations. The inclusion of essential amenities and facilities information enhances transparency and ensures that guests know what to expect during their stay.
(www.Travelodge.com, 2023)

- **Drawbacks**

The shortcoming identified in the Travelodge's website experience occasional technical glitches or errors due to its nature of managing highly complexed data and interface. These issues could range from slow loading times to error messages during the booking process. Such interruptions can disrupt the user experience and deter guests from completing their bookings or accessing important information.

- **About Booking.com**

It is a well-known online travel agency that specializes in accommodation reservations. It allows users to book hotels, apartments, hostels, and other types of accommodations worldwide. Users can search for properties based on various criteria such as location, price, facilities, and guest reviews. Booking.com offers a wide range of options to suit different budgets and preferences, and it provides a user-friendly platform for browsing, booking, and

managing reservations. Additionally, Booking.com often features special deals and promotions, making it a popular choice for travellers seeking affordable accommodations.

- **Pro's**

One notable advantage lies in its extensive database of properties worldwide, providing users with a vast selection of options to choose from. This wide range of choices caters to diverse preferences and budgets, ensuring that users can find accommodations tailored to their specific needs. Additionally, Booking.com user-friendly interface simplifies the booking process, allowing users to easily navigate through search results, filter options, and property details. The website provides comprehensive information about each property, including detailed descriptions, photos, and guest reviews, empowering users to make informed decisions. (www.Booking.com, 2019)

- **Drawback's**

While Booking.com offers numerous advantages, some issues are seen that users may encounter. One notable drawback is the sheer volume of options available, which can sometimes overwhelm users and make it challenging to narrow down their choices. Additionally, despite the website's extensive database, some users may find that certain destinations or property types are not adequately represented, limiting their options. Furthermore, Booking.com's reliance on user-generated content, such as guest reviews, may lead to inconsistencies in the quality and reliability of information, making it difficult for users to assess the true value of a property.

- **About Marriott Hotel**

Marriott International is a well-known hospitality company that operates a range of hotel brands worldwide. The Marriott hotel website serves as a central platform for guests to explore and book accommodations at Marriott properties. The website offers a user-friendly interface, allowing users to search for hotels based on various criteria such as location, dates, and amenities.

Users can view detailed information about each hotel, including room types, facilities, dining options, and nearby attractions. The website often features special offers, promotions, and exclusive deals for Marriott Bonvoy members, the company's loyalty program.

- **Pro's**

The Marriott hotel website offers a seamless and user-friendly interface, providing guests with easy navigation and access to essential information about Marriott properties. Detailed descriptions, photos, and amenities listings empower guests to make informed decisions when booking accommodations. Moreover, the website frequently features special offers and promotions, including exclusive deals for Marriott Bonvoy members, enhancing the overall value for guests. With its intuitive layout and comprehensive details, the

Marriott website ensures a convenient and satisfying booking experience for travellers. (Marriott Bonvoy, 2024)

- **Drawback's**

However, the limitations are identified in the website. Despite its user-friendly interface, guests may find that the customization options are somewhat limited, as they may not be able to specify detailed preferences during the booking process. Additionally, for Marriott Bonvoy members, managing reservations and accessing exclusive deals may sometimes be complex or cumbersome. Despite these drawbacks, the Marriott hotel website remains a valuable tool for travellers seeking quality accommodations with a reputable hospitality brand.

3.2.2 Planning

The project plan commenced with outlining the topic, a decision made in October. Subsequently, research was conducted on various products to establish a robust platform encompassing GUI and data management as mentioned above. The vital aspect of software design was approached thoroughly, following a structured methodology:

- **Wireframes:** The initial step involved developing wireframes, providing a skeletal outline of the software's interface. This enabled visualization of the layout and functionality, laying the foundation for subsequent design phases.
- **Visual Designing or Prototyping:** Building upon the wireframes, visual design elements and prototypes were crafted through figma. This phase focused on integrating branding elements, colour schemes, typography, and imagery to create a visually appealing and cohesive user interface.
- **Responsive Designing:** Ensuring the software's adaptability across various devices and screen sizes was prioritized through responsive design techniques. This facilitated a seamless user experience, irrespective of the device used.
- **Database Optimization:** Optimal database selection and configuration were crucial for efficient data management. Careful consideration was given to factors such as data structure, storage requirements, and scalability to ensure smooth operation of the software.
- **Functioning of the Software:** Implementing core functionalities and features was undertaken next, ensuring that the software met the specified requirements and provided a seamless user experience.
- **Testing:** Rigorous testing procedures were employed to identify and rectify any potential issues or bugs. This iterative process involved comprehensive testing of both functionality and user experience aspects to deliver a polished and reliable software product.

3.2.3 Design

By following the above methods a design was established on which the code is supposed to be based on.

During the design phase of the hotel management software, the team used new and creative ideas. They used advanced tools like Figma for designing and Canva for planning the layout whereas HTML, CSS and JavaScript along with PHP was employed for the coding part. This helped them make the software easy to use and nice to look at. They made sure their ideas turned into real design parts smoothly.

User Interface Design layout:

Figma, a collaborative design tool, served as the pillar of the UI design process. Its cloud-based nature facilitated real-time collaboration among team members (if required), allowing for efficient idea and iteration. The versatility of Figma enabled the creation of detailed design components, from layout structures to elements, all within a single platform. Utilizing Figma's robust features, the UI design with a visually remarkable interface that prioritized usability and style was generated or developed. (www.Figma.com,2024).

In the initial stages of design exploration, Canva emerged as a valuable tool for wireframing. Its intuitive interface provided a smooth workflow for detailing the software's layout. With Canva, the wireframe mock-ups were, enabling rapid visualization of design concepts. The simplicity of Canva's drag-and-drop interface allowed for easy collaboration and feedback incorporation, paving the way for seamless transition into the subsequent design phases. (www.Canva.com, 2024) The design generated was latterly coded in a professional manner using tools like visual studio code and required extensions which were installed within the environment.

Future design plans:

The hotel management software plans to introduce several basic enhancements to improve user experience and functionality. These enhancements include refining the user interface for a more intuitive navigation experience and implementing a feedback system to gather user suggestions and improve service quality. As the primary focus will be on optimizing the UI to facilitate more intuitive navigation for users. This will include careful assessment of the current layout and structure of the software and making strategic adjustments to improve the flow of information and streamline the user journey. By enhancing clarity and simplicity in design, users will find it easier to locate relevant information and perform desired actions, such as room bookings and reservation management.

Moreover, consistency in design elements across all pages will be prioritized to create a cohesive and unified user experience. This consistency extends to aspects such as colour schemes, typography, and button styles, ensuring a visually harmonious interface that reinforces brand identity and build's user trust.

The hotel management software could incorporate voice control functionality, allowing users to interact with the system using voice commands. This advancement would give a more seamless and convenient user experience, enabling users to navigate through the interface, make reservations, and access information simply by speaking. By integrating voice control technology, the software can offer an additional layer of accessibility and ease of use, catering to a

wider range of users, including those with disabilities or those who prefer handsfree interaction.

Furthermore, efforts will be made to enhance the responsiveness of the frontend to ensure seamless access and functionality across a wide range of devices and screen sizes. With the increasing prevalence of mobile devices, including smartphones and tablets, it is essential to optimize the frontend to adapt to different screen resolutions and orientations. By embracing responsive design principles, the software will deliver a consistent and optimal user experience regardless of the device used.

3.2.4 Content

As it is crucial to write high quality, original content that is relevant to your company and products. Everything from the quality of your content to the keywords used affects your site's ranking in search engine results. Keyword research can provide topic ideas and, if used sparingly, will help customers quickly locate the content they're seeking. To increase the appeal, each web page should have no less than 150 words and should include links to credible resources and other pages on your website. More than providing information, the content published on your site should be written to echo your brand's unique voice. Before publishing, use a tool like to ensure that it passes duplication standards. Posting duplicate content on your website will result in steep penalties from Google, including possible removal from search engine results.

As one of the world's leading suppliers of stock photos, Getty Images regulates the appropriate use of images online. If not posted with permission, images that are copyrighted may result in fines. When incorporating images into your site's content, make sure to receive written permission for use and pay relevant fees, if applicable. There are also sites that offer access to copyright-free images. However, be sure to carefully review terms of use before posting them. (Rand. S, 2014)

By carefully following the standards mentioned above, the data on the website was organized and arranged to ensure that it didn't copy others' work or violate copyrights. The webpage has also gone through a process of identifying plagiarism through a duplicate text identifier. (www.Copyscape.com, 2019)

The content displayed on the software undergoes manipulation to ensure its relevance and alignment with the software requirement as it portrays a company's branding and service offerings. Primarily featured on the home page and in terms and conditions, this content explaining the missions of project developed, values, and the range of services it provides to its clientele. To uphold a standard of excellence, certain portions of the content are curated from reputable sources within the hotel management industry, ensuring consistency and quality across the website.

The inclusion of compelling visuals such as images and videos further enrich the user experience. These multimedia elements are selected from diverse sources across the internet to complement the written content and engage visitors effectively. By integrating visually appealing imagery and dynamic videos, the

software aims to captivate the audience's attention and convey the essence of the company's offerings in a more immersive manner.

Furthermore, the strategic placement of these multimedia elements on the website contributes to enhancing its overall appearance and user engagement. Whether it's showcasing luxurious hotel amenities or highlighting the unique features of the company's services, each image and video is thoughtfully curated to resonate with the target audience and evoke a sense of desirability.

3.2.5 Functionality

PHP connects to the database, sends the query, and receives the results; but the query itself needs to be written in SQL. Although SQL is a common standard, there are many dialects of SQL. Each database vendor, including MySQL, has added extensions to the standard language. These improve efficiency and functionality but are usually incompatible with other databases.

PHP communicates with the database and stores the results, queries need to be written in SQL, the standard language used to query a relational database. (Powers, D. and Springerlink, 2010)

There are hundreds of programming languages to choose from, but there are a few things that make JavaScript special. Most important is its relationship with web browsers, such as Google Chrome, Safari, Microsoft Edge, and Firefox. Almost every web browser can run JavaScript, which means the code you write in JavaScript can be run on any computer with a web browser. There's no need to install any additional special software. Almost all smartphone web browsers can run JavaScript, too, so you may have a JavaScript-enabled browser in your pocket or bag right now.

Because of its relationship to web browsers, JavaScript is an extremely important part of web development. If a website includes dynamic, interactive features, it was probably created with JavaScript. For example, YouTube uses JavaScript to show you previews of videos when you hover over their thumbnails, Threads uses JavaScript to load more posts as you scroll down the page, and Amazon uses JavaScript to power its Look Inside feature. (Morgan, N. 2023).

In the software, once a user requests information, an integration process activates, tapping into the database to fetch relevant data. This process is handled through PHP scripting, a robust tool employed to retrieve and manipulate data stored in the database. For instance, when a customer initiates the hotel room booking process, PHP scripts swing into action, retrieving pertinent details such as available room types, pricing, and booking status from the database.

Furthermore, the system doesn't just stop at retrieving data; it also facilitates dynamic interactions to cater to various user needs. For example, users can modify their booking details, such as the duration of their stay or the type of room preferred or cancel their booking if no longer required directly through the system interface. Upon receiving these modifications, PHP scripts swiftly update the corresponding

records in the database, ensuring real-time synchronization between user inputs and system responses.

Moreover, the system highlights a robust search functionality that allows users to effortlessly browse through available rooms based on their preferences. Through intuitive user interfaces as the information of the rooms is made available to users through the database itself, users can specify their criteria in terms of type of room in their search queries. Behind the scenes, PHP scripts process these queries, executing database queries to retrieve matching room listings. This process ensures that users receive accurate and up-to-date search results tailored to their requirements. Additionally, the system incorporates secure payment processing functionality to streamline the booking process. Users can make payments securely through integrated payment gateways, confident in the system's adherence to stringent security protocols. PHP scripts orchestrate the payment process, handling transaction requests, verifying payment details, and updating payment status in the database upon successful transactions.

JavaScript plays a crucial role in executing basic functions on the website. Through JavaScript, interactive elements and content are facilitated, enhancing user engagement and usability. For instance, JavaScript enables features such as form validation, where input fields are checked for accuracy before submission, ensuring data integrity. Additionally, JavaScript is utilized to create interactive elements like dropdown menus, image sliders, interactive textboxes, allowing users to interact with the website effortlessly.

4. Methods & Development Approach

On analysing various types of approach to begin my development with. I stumbled upon a methodology called “Traditional Web Development or Waterfall methodology”.

4.1 The Waterfall Methodology

It is the simplest methodology of software development and the first Process Model, which was introduced in 1970. In the Waterfall methodology, the development process looks like a stream, successively passing through phases one by one. However, the transition from one phase to another occurs only after the completion of the previous one. In other words, each phase must be completed before the next phase can begin. The main disadvantage of this model is that a change in one of the phases will entail an inevitable change in all subsequent stages. It means that the Waterfall model is good only for short and easy projects. (www.adcisolutions.com, 2024)

Five phases comprise the Waterfall methodology: requirements, design, implementation, verification, and maintenance. Let's break down the five specific phases of Waterfall development and understand why it's critical to complete each phase before progressing to the next.

The requirements phase states what the system should do. At this stage, you determine the project's scope, from business obligations to user needs. This gives you overview of the entire project. The requirements should specify:

- resources required for the project.
- what each team member will work on and at what stage.
- a timeline for the entire project, outlining how long each stage will take.
- details on each stage of the process.

But these requirements may range from very abstract to a detailed mathematical specification, professor of computer science at Old Dominion University. That's because requirements might not outline an exact implementation, and that's something development addresses in later stages. (Atlassian, 2023)

The Waterfall methodology aligns perfectly with the project, which involves using HTML, CSS, JavaScript, and PHP for frontend design, functionality, and data storage. Moreover, HTML and CSS can be used to create the user interface, design layouts, and style elements to ensure user-friendly experience. JavaScript can then be employed to add interactivity and dynamic behaviour to the website, enhancing its functionality and user engagement.

The key step that are thoroughly followed during the implementation of the methodology are further listed:

- **Requirements Gathering:** Gathering detailed specifications for the hotel management software.
- **System Design:** Planning the system architecture and user interface (prototyping).
- **Implementation:** Coding the frontend (HTML/CSS/JavaScript) and backend (PHP) functionality.
- **Testing:** Conducting thorough testing to ensure functionality and reliability.
- **Deployment:** Deploying the software for users to access online.

As the data information gathering and system design is mentioned rest of the steps are followed are stated below.

Waterfall Methodology

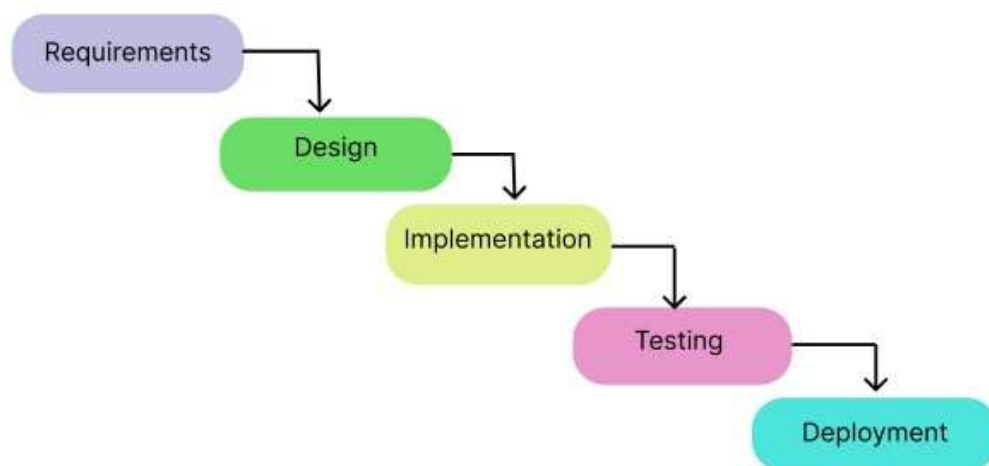


Figure-1

4.2. Languages Involved

HTML, CSS, JavaScript (JS), and PHP are fundamental languages in web development, each serving a specific purpose in creating dynamic and interactive websites. In the hotel management software project, these languages were employed for various reasons, aligning perfectly with the software's requirements and functionality.

HTML, or “Hypertext Markup Language,” has been around in some form since roughly 1993, when it was created by physicist Tim Berners-Lee. It’s now in its fifth generation, HTML5.

HTML isn’t a programming language, rather it’s a “Markup language.” It uses a syntax of tags to change the way text is displayed, for instance. It can also define where images go on a page. CSS or “Cascading Style Sheets” works with HTML to create the format for web pages. It works on top of the foundation of a page that was created with HTML. It also helps adapt pages to different formats optimized for desktops or mobile devices. The latest standard is CSS3, meaning they are in their third generation. CSS is what gives websites their polished and professional look.

JavaScript, which is often abbreviated to JS, is another front-end programming language that is one of the core technologies of the World Wide Web, alongside HTML and CSS. As of 2022, 98% of websites use JavaScript on the client side for webpage behaviour, often incorporating third-party libraries like jQuery. Whereas HTML & CSS are used to control presentation, formatting, and layout, JavaScript is used to control the behaviour of different web elements. (Knight. A, 2022)

PHP (short for Hypertext Preprocessor) is the most widely used open source and generalpurpose server-side scripting language used mainly in web development to create dynamic websites and applications. It was developed in 1994 by Rasmus Lerdorf. PHP can actually do anything related to server-side scripting or more popularly known as the backend of a website. For example, PHP can receive data from forms, generate dynamic page content, can work with databases, create sessions, send and receive cookies, send emails etc. There are also many hash functions available in PHP to encrypt user’s data that makes PHP secure and reliable to be used as a server-side scripting language. (GeeksforGeeks, 2020).

Alignment with the software

In the hotel management software project, the utilization of HTML, CSS, JavaScript (JS), and PHP aligns perfectly with the software's requirements and functionality, considering its fundamental nature and the project's scope. HTML's role in defining the structure of web pages and specifying content placement directly addresses the need for clear and organized presentation of hotel management services and information. CSS complements HTML by enhancing the visual appeal of the software, ensuring a polished and professional look that aligns with industry standards.

JavaScript's inclusion enables the implementation of interactive elements and dynamic behaviour, enhancing user engagement and providing a more intuitive browsing experience. Given the basic level of the software, JavaScript's capabilities can be used to introduce simple yet effective features such as form validation and interactive elements, contributing to the overall functionality of the application.

Additionally, PHP serves as the backbone of the software's backend operations, facilitating dynamic content generation, data processing, and interaction with databases. Despite the student's basic proficiency level, PHP's simplicity and versatility enable them to handle

essential server-side tasks effectively, such as receiving data from forms, generating dynamic page content, and working with databases. This aligns perfectly with the project's objective of developing a functional hotel management system that meets basic operational requirements while providing a valuable learning experience for the student developer.

4.3. Software and tools for website creation

To deploy the languages a robust tool and a single platform was required therefore the tool that was used for the building of the software and was responsible for frontend and backend coding is Visual Studio Code (VS code) as it was opted in accordance of my expertise in it.

Architecturally, Visual Studio Code combines the best of web, native, and language-specific technologies. Using Electron, VS Code combines web technologies such as JavaScript and Node.js with the speed and flexibility of native apps. VS Code uses a newer, faster version of the same industrial-strength HTML-based editor that has powered the "Monaco" cloud editor, Internet Explorer's F12 Tools, and other projects. Moreover, Visual Studio Code includes builtin support for IntelliSense code completion, rich semantic code understanding and navigation, and code refactoring. (Microsoft, 2016)

Visual Studio Code (VS Code) proved to be an essential tool in the development of the hotel management software project. Its versatility and extensive range of features significantly contributed to HTML, PHP and all the coding process. With its user-friendly interface and customizable layout, developers could efficiently write, edit, and debug code, thereby enhancing productivity and reducing development time. Moreover, the vast library of extensions available for VS Code provided specialized support for PHP development, enabling seamless integration with frameworks and libraries essential for the project's backend functionality.

phpMyAdmin is an open-source tool built on PHP that enables you to administer MySQL and MariaDB databases online. To use it, you'll need to install the software on a server running either Windows or one of the several Linux distros it supports. The software enables you to manage as many databases as you want. You can edit tables and values, create and delete databases, or even import and export them. The application itself is easy for beginners to pick up, but it offers enough depth that it can take a while for you to master everything it offers. (Elegant Themes, 2022).

PhpMyAdmin played a crucial role in database management throughout the project. As a webbased application, PhpMyAdmin provided a user-friendly interface for administering MySQL databases, allowing developers to create, modify, and manage database structures effortlessly. Its intuitive interface facilitated tasks such as importing and exporting data, executing SQL queries, and optimizing database performance. By simplifying database management tasks, PhpMyAdmin enhanced the efficiency of the development process and ensured the smooth functioning of the hotel management software's backend.

XAMPP, which stands for Cross-Platform, Apache, MySQL, PHP, and Perl, is a free platform that allows developers to test their code locally on their own computers. This platform provides the experience of having your own mini web server at home, compatible with both Windows (WAMP) and Linux (LAMP) environments. (edX, 2023).

In the development of the hotel management software project, phpMyAdmin was chosen as the database management system. However, phpMyAdmin requires a server to run on, and among the options available like USBWebserver, the decision was made to utilize XAMPP. Serving as

the local server environment, XAMPP played a crucial role in facilitating testing and deployment of the application locally. It allowed developers to simulate a production environment on their local machines, ensuring thorough testing of the software's functionality before deployment. This choice ensured consistency between the development and production environments, reducing the risk of compatibility issues and optimizing the development workflow. Additionally, XAMPP provided essential tools such as Apache, MySQL, and PHP, laying a solid foundation for the project's backend infrastructure.

4.4. SQL/database platform

The database development process followed a lifecycle concept, which emerged during the research phase while exploring various database development methods and structures. As I delved into different approaches to database design and architecture, the concept of a lifecycle approach became apparent. Consequently, the forthcoming database structure was defined with this concept in mind.

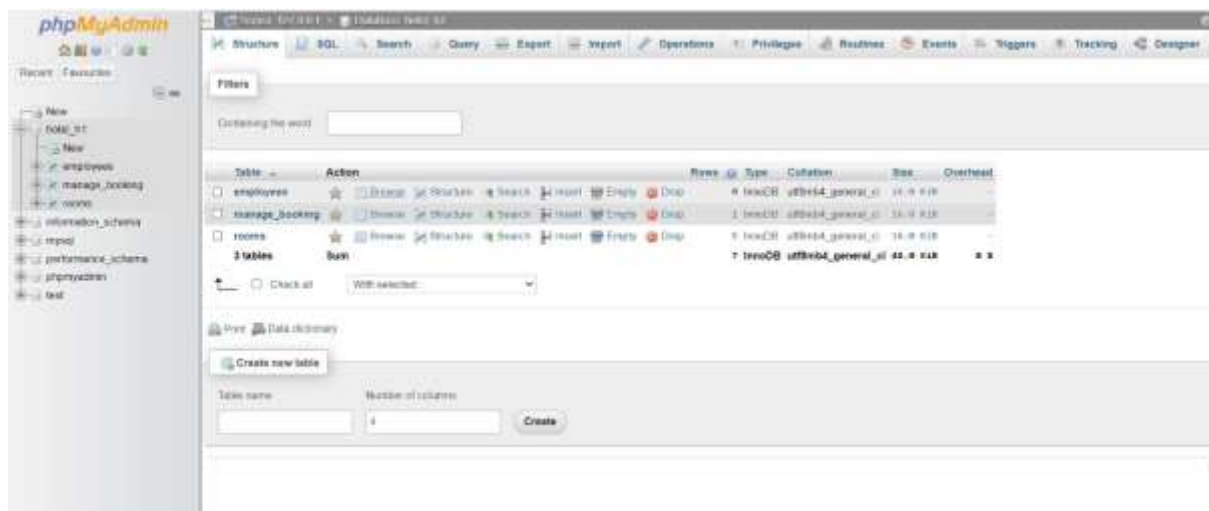


Figure-2

There are various stages in database development. However, it is not necessary to follow each of the steps sequentially. The life cycle can be divided into three phases:

4.4.1 Define the objective of your database:

The first step is to determine the purpose of your database. For example, if you are running a small home-based business, you can design a customer database that maintains a list of consumer info to generate emails and reports. Hence, understanding the importance of a database is vital.

4.4.2 Locate and consolidate the necessary data.

The next step is to collect all kinds of information you might want to store in the database. Begin with the existing data and mull over the questions you want your database to answer. It will help you decide which data needs to be recorded.

4.4.3 Distribute the data into tables:

Once you have amassed all the necessary data items, the next step is to divide them into main entities or subject areas. For example, if you are a retailer, some of your

main entities could be products, customers, suppliers, and orders. Each entity will then become a separate table.

4.4.4 Change data items into columns:

Data is segregated into tables, such that every data item becomes a field and is shown as a column. For instance, a customer table might include fields like name, address, email address, and city.

4.4.5 Identify primary keys:

The next step to improve your database design is to select a primary key for every table. This primary key is a column or a set of columns that pinpoint each row distinctively. For instance, in your customer table, the primary key could be customer ID. It will allow you to identify unique rows based on the customer ID.

(Naeem. T, 2019)

The primary purpose of the database development lifecycle within the hotel management software project is to efficiently manage and organize various aspects of hotel operations, catering to both staff and guest needs. This encompasses tasks such as managing reservations, guest profiles, room availability, staff details, and other essential information pertinent to hotel management. By establishing a wellstructured database, the software aims to ease hotel operations, enhance guest experiences, and facilitate smoother interactions between staff and guests.

In the database, diverse types of data are stored to support the functionality of the hotel management software. This includes guest information such as names, contact details, booking history, and preferences, as well as room availability, rates, reservation dates, and staff schedules. Additionally, employee data such as names, roles, contact information, and work schedules are also stored to facilitate efficient staffing and management within the hotel.

Through intuitive user interface (UI) and user experience (UX) design, the data stored in the database is divided into tables, reflecting logical entities within the hotel management system. For instance, the UI/UX design may include separate sections for guests, rooms, reservations, and staff, with each section corresponding to a distinct table in the database. This division ensures that data is organized in a structured manner, facilitating easy navigation and access for users.

As data is initialized within the tables, fields corresponding to each data item are created, defining the structure of the database. For example, within the "Customers" table, fields such as customer ID, name, contact details, and booking history are initialized to store relevant guest information. Similarly, the "Employees" table includes fields such as employee ID, name, role, and contact details to manage staffrelated data. This initialization process ensures that the database is properly configured to store and manage the required information effectively.

Furthermore, in identifying potential primary keys (PK), both the "Customers" and "Employees" tables designate unique identifiers (IDs) as primary keys. These IDs serve as unique identifiers for each record within the respective tables, allowing for efficient retrieval and manipulation of data. By establishing primary keys, the

database ensures data integrity and facilitates seamless operations within the hotel management software.

4.5. Design approach for hotel website.

The design approach was formulated subsequent to the completion of the Figma and Canva models, serving as pivotal tools for prototyping and wireframing. These platforms facilitated the visualization of several pages earmarked for implementation, strategically planned through their prototypes. The Home page held chief significance, serving with its navigation links to secondary pages, comprehensive hotel information, and access to site terms and conditions. Notably, it was preserved as `index.php`, signifying its role as the initial landing page upon executing the code.

In the development process of the hotel management software, eight distinct pages were carefully crafted to ensure smooth functionality and user satisfaction. At the forefront stands the Homepage, a critical element offering users a comprehensive overview of the hotel's services, alongside essential navigation links and access to pertinent information such as terms and conditions. Following this, the Search Room page was designed to facilitate easy exploration of available accommodations based on diverse criteria like dates, occupancy, and amenities. Upon room selection, users are smoothly transitioned to the Payment Page, where they finalize their reservations securely. Complementing these user-centric features, the Manage Booking page empowers users to review, modify, or cancel existing reservations, fostering a sense of control over their bookings.

Moreover, the software incorporates an Employee Login feature embedded across all pages, granting authorized personnel access to the Employee Dashboard—a centralized platform for managing bookings, guest inquiries, and administrative tasks. Additionally, employees can easily navigate to the View Bookings page, facilitating efficient monitoring and management of guest reservations. Through the thoughtful design and implementation of these pages, the software aims to streamline the user experience and optimize hotel management operations. All these options and pages were created using Figma as the primary prototyping tool, allowing for intuitive navigation and efficient functionality. The wireframes for each page were developed using Canva, a user-friendly graphic design platform known for its accessibility and efficiency. This combination of tools enabled the translation of conceptual ideas into tangible digital experiences, paving the way for an effective and user-friendly hotel management system.

4.6 Wire Frames Developed

Login

Figure-3

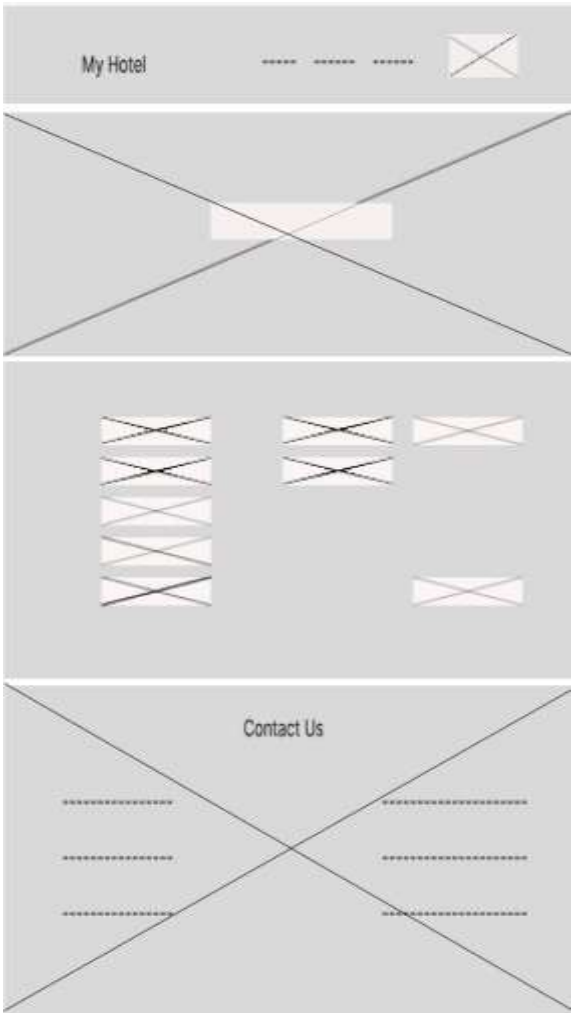
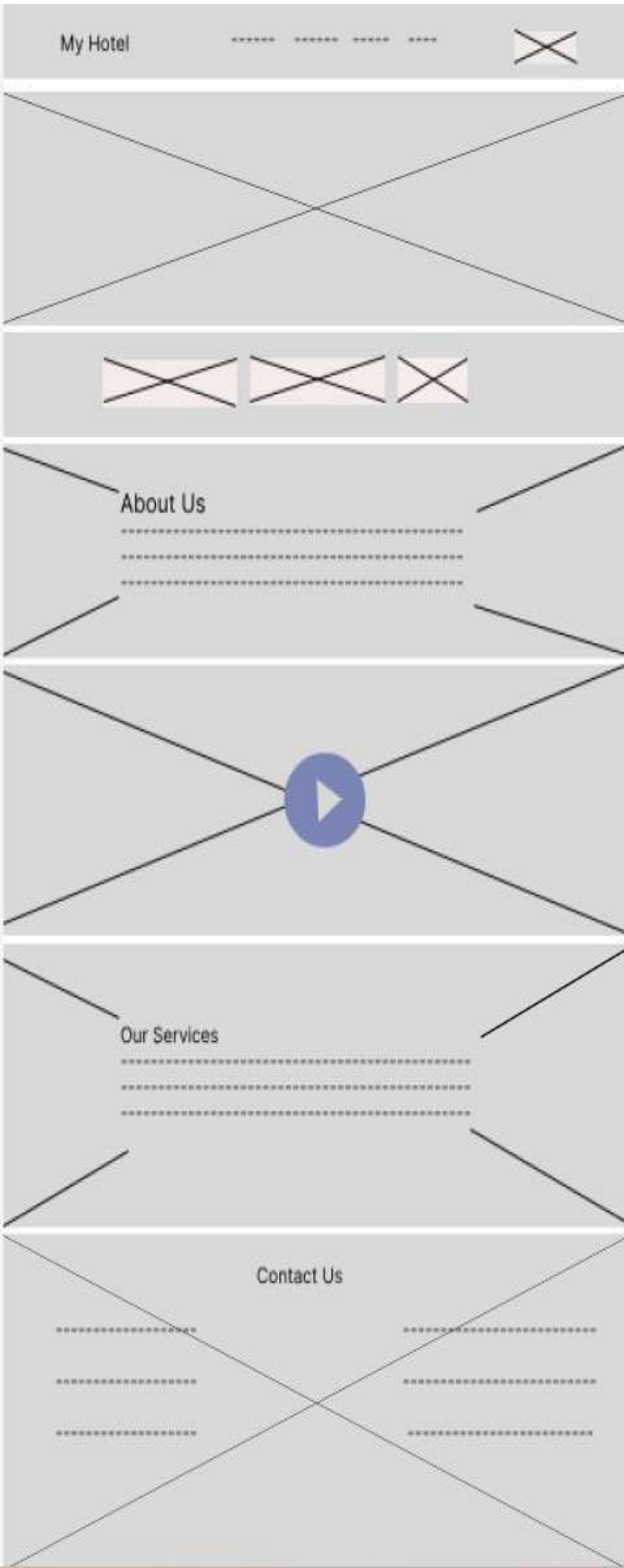


Figure-4

Figure-5

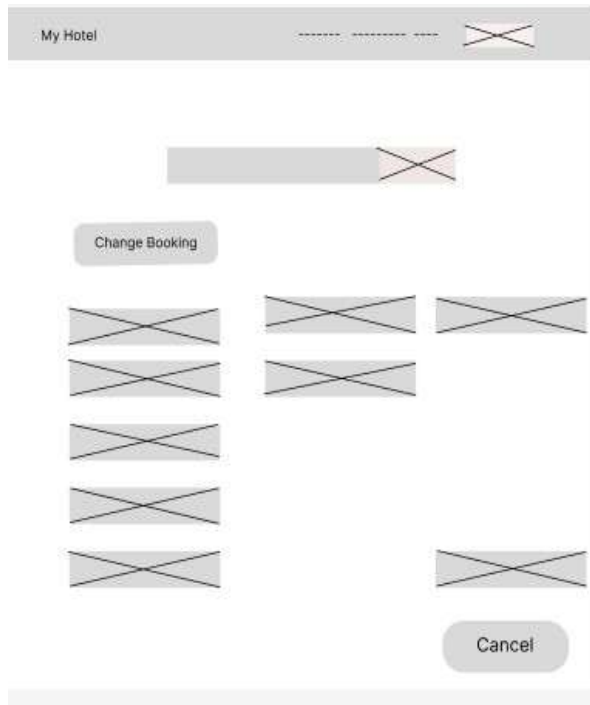


Figure-6

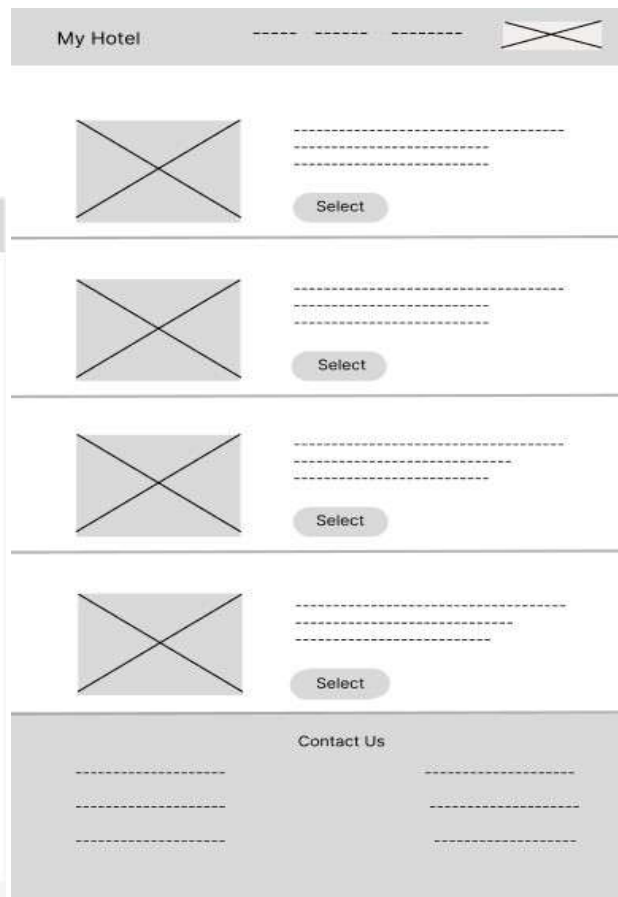


Figure-7

4.7 Prototypes Developed



Figure-9

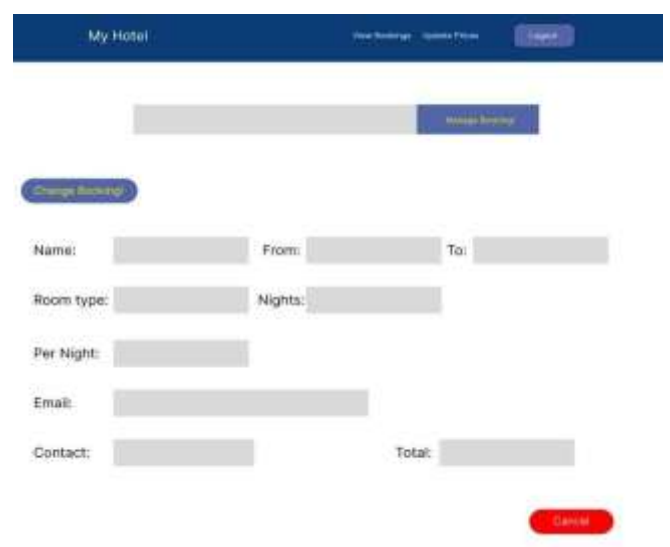


Figure-10

My Hotel

Home Contact Us About Us Login

Manage Booking

Name: From: To:

Room type: Nights:

Per Night:

Email:

Contact: Total:

Change Booking! Cancel

Contact Us

Phone no: +44 (0) 7566235125

Email: myhotel1@mail.co.uk

Instagram: my_Hotel_reviews

Privacy Policy

Terms and Conditions

Report an Issue

Figure-11

5. Implementation

5.1. Visual Studio code Setup

As for the software development it was the most crucial tool that was utilized during the development process though I did face some complications during the installation process and made it work. The steps followed for downloading are listed below:

Step 1: Open your web browser and go to the Visual Studio Code website.

Step 2: Download the appropriate version for your operating system (e.g., Windows).

Step 3: Install the software by agreeing to the license agreement and following the prompts.

Step 4: After installation, ensure to launch Visual Studio Code by selecting the appropriate checkbox.

Step 5: Explore and download extensions to customize your coding experience and optimize your workflow.

(GeeksforGeeks, 2022)

The steps outlined above were followed to install Visual Studio Code, ensuring a smooth setup process. Upon completion, available extensions were explored and downloaded several crucial ones to enhance the coding experience. These extensions, including Live Server, PHP Server, IntelliSense, and those for HTML, CSS, and PHP, play a pivotal role in optimizing the workflow and providing additional functionalities tailored to my programming needs. IntelliSense: This extension provides intelligent code completion, code snippets, and parameter info, significantly improving coding efficiency and accuracy. With IntelliSense, I can quickly access relevant code suggestions and autocomplete features, reducing errors and enhancing productivity.

Live Server: Live Server is indispensable for web development projects as it enables real-time browser refreshing and live editing capabilities. By automatically reloading web pages upon saving changes to code files, Live Server allows for seamless testing and debugging of web applications without the need for manual refreshes.

PHP Server: As a backend developer, PHP Server is essential for local PHP development and testing. This extension facilitates running PHP scripts locally on my machine, replicating the server environment and enabling rapid prototyping and debugging of PHP-based applications.

HTML CSS Support: This extension provides advanced support for HTML and CSS languages, including syntax highlighting, code formatting, and autocomplete. With HTML CSS Support, I can write clean and organized code more efficiently, ensuring consistency and readability across web development projects.

PHP IntelliSense: Building upon the capabilities of IntelliSense, PHP IntelliSense offers intelligent code completion and documentation for PHP files. This extension enhances PHP development by providing context-aware suggestions and detailed documentation for PHP functions, classes, and variables, facilitating faster coding, and reducing errors.

A challenge was encountered when attempting to use Live Server in Visual Studio Code to review the HTML output. However, after watching a tutorial on YouTube, it was discovered that Live Server could be activated simply by right-clicking, enabling seamless viewing of the output. Additionally, another hurdle arose when it was realized that PHP needed to be downloaded externally and the path set in the system's environmental variables for both PHP and XAMPP servers to function properly. Despite these initial setbacks, the issues were successfully resolved by seeking guidance from the official website of the XAMPP server, ultimately overcoming the obstacles with newfound knowledge and confidence.

5.2 Database Initialising and Table creation

The database used is phpMyAdmin as mentioned above it was leveraged using xampp server. Few steps were followed to gain access stated below:

Step 1: Download and install XAMPP.

Begin by installing the XAMPP program. It's simple to download the required files from the official website of the open-source Apache distribution.

Step 2: Start web server and database in XAMPP.

You can start the program once the XAMPP download is complete. Start the Apache web server and the MySQL database in the XAMPP control panel by clicking on the 'Start' button to use phpMyAdmin.

Step 3: Open the web interface of phpMyAdmin.

You can easily open the phpMyAdmin web interface in your browser. Simply type the address 'http://localhost/phpMyAdmin' into the browser's search bar. This will automatically bring you to the phpMyAdmin start page. You can adjust the settings and create or manage your own databases here. (IONOS Digital Guide, 2023)

Following the guidance provided, the steps outlined for installing and configuring XAMPP were diligently followed to set up the required environment for the hotel management software. Upon successful installation, the Apache web server and MySQL database were started using the XAMPP control panel, thereby enabling access to phpMyAdmin for database management. Subsequently, a database named "My_hotel" was created within phpMyAdmin, serving as the cornerstone for storing crucial data related to employee details, room information, and guest bookings.

Within the "My_hotel" database, three essential tables were crafted to organize and manage data efficiently. The "employee" table was structured to capture pertinent information about hotel staff, including their unique EmployeeID, full name, contact details, email address, salary, and position. Each attribute within the table was carefully selected to ensure comprehensive coverage of employee data, facilitating seamless management of staff records within the software.

Similarly, the "Rooms" table was designed to store detailed information about the available accommodations offered by the hotel. Attributes such as Room_ID, Price, Facility, Type, ImagePath, and others were meticulously defined to provide a comprehensive overview of each room's features and amenities. This structured approach ensures that guests can make informed decisions when booking rooms through the software.

Lastly, the "manage_booking" table was crafted to facilitate the management of guest bookings efficiently. Attributes such as Booking_ID, Name, Room_Type, Per_Night rate, Email, Contact, Nights, From_Date, and To_Date were carefully chosen to capture essential booking details, including guest information, room preferences, duration of stay, and booking dates. This organized framework allows hotel staff to effectively manage guest reservations, monitor room availability, and streamline the check-in/check-out process.

Future Plans for the database

To further enhance the functionality of the hotel management software database, additional tables can be incorporated to capture supplementary information relevant to hotel operations. One potential addition is a "Services" table, which can store details about the various amenities and services offered by the hotel, such as room service, laundry service, spa treatments, etc. This table can include attributes such as Service_ID, Service_Name, Description, Price, and Availability to facilitate efficient management and booking of hotel services.

Another beneficial addition could be a "Feedback" table, dedicated to collecting guest feedback and reviews about their stay at the hotel. This table can include attributes such as Feedback_ID, Guest_Name, Date, Rating, Comments, and Room_Number to capture valuable insights into

guest satisfaction and areas for improvement. Analyzing this feedback can help the hotel management team identify trends, address concerns, and enhance the overall guest experience.

5.3. Developing Front-end and backend (PHP)

In this section, we'll delve into the development of the front-end and backend components of the hotel management system. This system aims to provide users with a platform to manage bookings, view available rooms, and perform various administrative tasks. The development process involves creating nine distinct pages, each serving a specific purpose and utilizing both front-end HTML/CSS and backend PHP scripting.

5.3.1 Homepage

The `home.php` page serves as the welcoming gateway to the hotel's website, offering visitors a glimpse into what the organization has to offer. It functions as the primary landing page, presenting users with an introduction to the hotel's amenities, features, and possibly promotional offers. Navigation is facilitated through a user-friendly interface, featuring a top navigation bar allowing seamless exploration of other sections, such as booking management, hotel information, and customer support. The inclusion of a login button hints at the potential for user authentication to access employee dashboard and other relevant services. It also includes a button "Rooms" which redirects to rooms page from where customers are allowed to place their booking.

Implemented with a blend of HTML and CSS, the page's HTML structure likely includes elements like headings, paragraphs, and navigation links. Meanwhile, CSS stylesheets enhance the visual appeal by defining fonts, colors, layouts, and backgrounds. Though specifics of the CSS code are not provided, typical styling may involve setting background imagery, creating sticky navigation bars, ensuring consistent typography, and optimizing for various devices through responsive design principles. Overall, `home.php` functions as an inviting entry point to the hotel's online presence, marrying functionality with aesthetic appeal to captivate and engage users effectively.

5.3.2 Rooms Page

The `"rooms.php"` page serves the purpose of displaying available rooms along with their details such as price, facilities, and room type. The implementation of HTML in this page is primarily focused on structuring the content to ensure clarity and ease of navigation for users. Each room listing is encapsulated within HTML elements, with appropriate classes and IDs assigned for styling and scripting purposes. The HTML markup includes elements like `<div>`, `<img>`, `<table>`, and `<button>`, each serving a specific purpose in presenting the room information.

Through carefully crafted CSS styles, the room listings, including images, pricing information, and buttons, were designed and structured. CSS was utilized to define the layout, colors, and spacing, ensuring consistency and coherence throughout the page. By applying CSS rules to elements such as `divs`, `images`, `tables`, and `buttons`, the presentation of room details was optimized for different screen sizes and

devices, creating a well build browsing experience for users. Additionally, CSS was instrumental in creating visually appealing transitions, hover effects, and animations to engage users and highlight important information, contributing to an immersive browsing experience on the "rooms.php" page.

PHP is implemented in the "rooms.php" page to dynamically fetch room data from a database and populate the HTML content with the retrieved information. This dynamic data retrieval ensures that the room listings are always up-to-date and reflect the latest information stored in the database. In the provided PHP code snippet, a database query is executed to retrieve room details such as price, facility, room type, and image path from the database. The fetched data is then used within a PHP loop to generate HTML markup for each room listing dynamically. This approach allows for scalability and easy maintenance as new rooms can be added to the database without the need to modify the HTML markup manually. Additionally, PHP facilitates the integration of business logic, such as handling user interactions and processing form submissions, if applicable.

Moreover, on selecting the room and dates of stay the user is guided to the payment page where he is required to input his details including the card details for the purpose of booking.

5.3.3 Payment Page

The "payment.php" page serves the purpose of facilitating the payment process for users who have selected a room and are ready to proceed with booking. The HTML implementation on this page primarily focuses on creating a structured form that collects essential information about users and the payment information. The HTML form includes input fields for the user's full name, card number, expiry date, CVV, and possibly other relevant details. Additionally, buttons are included to allow users to submit their payment details and proceed with the transaction.

CSS have been utilized to adjust the size, alignment, and spacing of input fields, ensuring they are visually distinct and easy to interact with. Additionally, CSS styles have been applied to buttons to make them prominent and visually appealing, encouraging users to complete the payment process. Furthermore, CSS is an instrumental tool in creating a responsive design, ensuring that the payment form is accessible and functional across different devices and screen sizes.

5.3.4 Manage Booking Page (Guests)

The "managebooking.php" page serves the purpose of allowing customers to view and manage their bookings. HTML is implemented to structure the content, including booking details such as the booking ID, customer name, room type, dates, and contact information. Each booking entry is encapsulated within HTML elements, making it easy for customers to navigate and interact with their bookings.

CSS was responsible in designing the layout and appearance of elements on the "managebooking.php" page. It was used to style textboxes and buttons, ensuring they stand out against the background and providing visual feedback to users. By customizing the appearance of elements with CSS, the page achieved an attractive and user-friendly design, enhancing user engagement and navigation.

PHP is utilized to retrieve booking information from a database and populate the HTML content with the relevant data. This data retrieval ensures that customers see up-to-date information about their bookings. Additionally, PHP may be used to implement functionality such as allowing customers to update or cancel their bookings directly from the page. While this page is intended for customers and not employees, PHP ensures that customers have access to accurate and personalized booking details, enhancing their overall experience with the booking system.

5.3.5 Login Page

The purpose of the "login.php" page is to provide users with a platform to authenticate themselves by entering their credentials, typically a username and password. The HTML implementation on this page focuses on creating a structured form where users can input their login details. HTML elements such as `<input>` fields and `<button>` elements are utilized to construct the login form, allowing users to enter their username and password and submit the form for validation.

CSS was instrumental in designing both the login box and the background of the "login.php" page. It styled the input fields, buttons, and elements of the login box, creating a sleek and modern appearance. Additionally, CSS customized the background colour, image, or pattern, enhancing the overall outlook and complementing the login form. Overall, CSS build the login page more visually appealing and user-friendly login interface.

PHP implemented on the "login.php" page is to handle the authentication process and validate user credentials against a database or other authentication system. It is responsible for processing the form submission, retrieving the entered username and password, and verifying them against the stored credentials. The implementation of PHP adds functionality to the login page by enabling secure user authentication and access control, contributing to the overall security and usability of the application.

5.3.6 Manage Booking Page (Employees)

The "afterlogin.php" page serves as a dedicated interface for employees who have successfully logged into the system. In terms of HTML implementation, the page structures its content to provide a user-friendly dashboard experience. This involves organizing elements such as navigation menus, informational panels, and data tables to facilitate efficient task management for employees. HTML is responsible for defining the layout and structure of the page, ensuring clarity and ease of use for employees navigating various features and functionalities.

CSS is utilized to design and style interactive elements such as textboxes, buttons, and navigation links. For example, CSS rules may be used to define the size, shape, colour, and hover effects of textboxes, ensuring they blend seamlessly with the overall design scheme of the page.

PHP is implemented on the "afterlogin.php" page to provide functionality and interaction with server-side resources. This includes fetching and displaying personalized employee information, such as their name, role, or recent activities. PHP scripts may also handle data processing tasks, such as querying a database for specific records or updating employee-related data based on user input. The use of PHP enables the page to deliver dynamic content tailored to each logged-in employee, enhancing the user experience and providing valuable insights and functionality for managing tasks efficiently. Additionally, it facilitates session management to authenticate users and control access to certain features or data based on their assigned roles and permissions within the organization. Overall, it enhances the functionality and security of the "afterlogin.php" page, making it an essential tool for employees to access and manage their tasks effectively.

5.3.7 View Bookings Page

The "viewbooking.php" page serves the purpose of allowing authorized employees to view all bookings made at the hotel. HTML is implemented on this page to structure the content and create a user-friendly interface for displaying booking details. The HTML markup includes elements such as tables to organize the booking information effectively, headings to provide clarity, and links for navigation purposes.

CSS is utilized to style various aspects of the table, including borders, background colour, font styles, and alignment. CSS rules define the border properties, background colour, and font styles of the table, ensuring readability and consistency. Additionally, CSS aligns the content within the table cells for a neat and organized presentation.

PHP is implemented in the "viewbooking.php" page to retrieve booking data from the database and populate the HTML content or the table present with the fetched information. Authorized employees can view the bookings stored in the database by executing PHP code that performs database queries to fetch relevant data. The PHP code then processes the retrieved data and generates HTML markup to display the booking details on the page in the tabular form.

5.3.8 Connection file (conn.php)

The "conn.php" file is responsible for establishing a database connection to the MySQL database server. It typically contains PHP code that uses the `mysqli_connect()` function to connect to the database server, specifying parameters such as the host, username, password, and database name. Once the connection is established, it sets the character set to UTF-8 to support international characters. This file ensures that other PHP scripts requiring database access can include it to establish a connection without duplicating connection code across multiple files, promoting code reusability and maintainability.

5.4. Refinement and finalization

Following the initial development phase, the software underwent a rigorous refinement and finalization process to ensure its readiness for deployment and use. This stage involved

comprehensive testing, debugging, and optimization to address any remaining issues and enhance overall performance and usability explained in the section below.

6. Analysis/Evaluation

6.1. Usability testing

Usability testing, as defined by Ginny Redish, is a method used to evaluate a product by testing it with representative users. The purpose of usability testing is to identify any usability problems, collect qualitative and quantitative data, and understand user behaviour, preferences, and satisfaction with the product. It involves observing users as they interact with the product to perform specific tasks, gather feedback on their experiences, and uncover areas for improvement. Usability testing helps ensure that a product is easy to use, efficient, and meets the needs and expectations of its intended users. (Redish. G, 2014)

About Ginny Reddish

Ginny Redish is a renowned expert in the field of usability and user experience, known for her contributions to user-centered design methodologies and writing influential books such as "Letting Go of the Words."

In 1979, Ginny founded the Document Design Center at the American Institutes for Research in Washington, DC, which she directed for 10 years. In 1985, Ginny set up one of the first independent usability test laboratories in North America. Users came to try out interfaces and documentation from Hewlett-Packard, IBM, SAP, Sony, and other clients. As President of Redish & Associates, Inc., over 25 years, Ginny helped clients in more than 400 projects. Ginny is a graduate of Bryn Mawr College and has a Ph.D. in Linguistics from Harvard University. (Reddish, 2023).

The primary steps guided by Ginny reddish for usability testing are:

- **Planning** • **Conducting** • **Analysing** • **Reporting**

6.1.1 Planning

In the planning stage of usability testing for the hotel software, objectives encompassed evaluating the software's functionality, navigation flow, and visual presentation to ensure a seamless user experience. As the sole participant, the testing scope extended to validating the functionality of navigation buttons and confirming successful database connections, aligning with the software's core features. Leveraging scenario-based testing, simulations were conducted to emulate the experiences of both guests and employees, ensuring comprehensive coverage of user interactions. The defined timeline facilitated structured testing, accommodating thorough examination of all software components, including booking management, room selection, and administrative functionalities.

6.1.2 Conducting

During the conducting stage, usability tests involved thorough navigation across the entire website to evaluate the functionality of navigation buttons and features. The connection page was accessed to display PHP information, confirming the successful establishment of the database connection. Additionally, colour combinations were tested to ensure optimal visibility and readability for users, enhancing the overall user experience. Scenario-based testing simulated the experiences of both guests and

employees, ensuring all functions, such as booking rooms, managing bookings, and accessing employee-specific features, were functioning as intended. Furthermore, the insertion and retrieval of data through the forms created were tested to ensure seamless interaction and data processing. This involved entering sample data into input fields and submitting the forms to verify that data was successfully stored in the database and retrieved accurately when requested, validating the integrity of the software's data management capabilities.

6.1.3 Analysing

During the analysis phase, the software's performance was evaluated, revealing seamless and quick redirection to pages with no observed issues. The chosen colour combinations and font styles contributed to optimal visibility, enhancing the overall user experience. Scenario testing for employees involved tasks such as logging in, viewing bookings, and managing them based on customer requests, all of which were executed seamlessly, leveraging database functionality. However, as a guest, it was noted that while navigating from the room selection to the payment details page was smooth, the secure payment feature requires authentication, hindering its functionality. Additionally, the system accurately calculated the number of nights based on selected dates but displayed an error prompt when the dates were entered out of order, highlighting a need for improved user guidance in handling date inputs. Also, it was observed that the view booking page efficiently retrieved data, providing quick access to booking information. This swift retrieval process facilitated seamless browsing for authorized employees, enabling them to promptly access and manage booking details without experiencing significant delays. The responsiveness of the view booking page contributed to a smooth user experience, enhancing productivity and facilitating efficient management of guest reservations.

6.1.4 Reporting

6.1.4.1 Functionality and Navigation

The software demonstrated robust functionality, with navigation buttons and features working seamlessly.

Scenario-based testing confirmed that core functions such as booking rooms, managing bookings, and accessing employee-specific features were functioning as intended. However, the secure payment feature requires authentication, hindering its usability for guests. It is recommended to streamline the authentication process to ensure a smooth payment experience.

6.1.4.2 Visual Presentation

The chosen colour combinations and font styles contributed to optimal visibility and readability, enhancing the overall user experience.

The software accurately calculated the number of nights based on selected dates but displayed an error prompt when dates were entered out of order. Improvements in guiding users in handling date inputs are recommended to enhance usability.

6.1.4.3 Recommendations

Streamline the authentication process for the secure payment feature to improve usability for guests.

Provide clearer guidance for users when handling date inputs to prevent errors and enhance usability.

Continue to monitor and optimize data retrieval processes to maintain efficient performance, particularly for high-volume data queries.

Consider implementing additional features or enhancements based on user feedback to further improve the overall user experience and meet evolving user needs.

7. Conclusions & Recommendations

In conclusion, the hotel management software project has successfully achieved its objectives through the strategic utilization of various tools, meticulous database design, and a user-centric approach to interface design. Through comprehensive testing and evaluation, valuable insights have been gained to guide future enhancements and ensure continued optimization of the system. Moving forward, the project stands poised to deliver even greater value to both guests and employees in the hospitality industry.

7.1 Effective Utilization of Tools and Technologies

The project leveraged a variety of tools and technologies such as PHP, MySQL, phpMyAdmin, XAMPP, Visual Studio Code, Figma, and Canva to develop a robust hotel management system. These tools enabled efficient development, testing, and deployment of both front-end and back-end components, ensuring a smooth and userfriendly experience for both guests and employees.

7.2 Structured Database Design

The project adopted a structured approach to database design, with a clear focus on organizing and managing various aspects of hotel operations. The use of phpMyAdmin and XAMPP facilitated the creation of well-defined tables for storing guest information, room details, employee data, and booking records. This structured database design enhanced data integrity, scalability, and overall system performance.

7.3 User-Centric Interface Design

The user interface (UI) design was carefully crafted to provide an intuitive and engaging experience for both guests and employees. Pages such as the homepage, room selection, payment, and booking management were designed with usability and functionality in mind, allowing users to easily navigate, book rooms, and manage reservations with minimal effort.

7.4 Thorough Testing and Evaluation

Usability testing, guided by Ginny Reddish's methodology, was conducted to evaluate the software's functionality, navigation flow, and visual presentation. The testing process identified areas for improvement, such as streamlining the authentication process for secure payments and providing clearer guidance for handling date inputs. These insights from usability testing will inform future iterations of the software, ensuring ongoing enhancements to user experience and satisfaction.

7.5 Future Enhancements and Recommendations

The report concludes with recommendations for future enhancements, including refining the secure payment process, improving user guidance for handling date inputs, and continuing to optimize data retrieval processes. Additionally, suggestions are made to consider implementing additional features based on user feedback to further enhance the overall user experience and meet evolving user needs.

In summary, the hotel management software project successfully achieved its objectives of developing a user-friendly, efficient, and scalable system for managing hotel operations. By leveraging the right tools, adopting a structured database design, and prioritizing user experience, the project lays a solid foundation for future enhancements and continued success in the hospitality industry.

8. List of references

Elphick, D. (2024). Hotel Management: Meaning, Types, and Operations. [online] SiteMinder. Available at: <https://www.siteminder.com/r/hotel-managementstrategies/#what-is-hotel-management-> [Accessed 19 Mar. 2024].

Martijn (2019). Hospitality Industry; Everything You Need to Know About Hospitality. [online] Revfine.com. Available at: <https://www.revfine.com/hospitality-industry/#what-is-the-hospitality-industry>.

Editor, F.H.A. (2024). Introduction to the Hotel Industry: The Essential Guide. [online] FHA. Available at: <https://fhahoreca.com/blog/hotel-industry>. [Accessed 19 Mar. 2024].

Kurtz, J. (2022). What Is the Hospitality Industry? [online] WebstaurantStore. Available at: <https://www.webstaurantstore.com/blog/3726/what-is-the-hospitality-industry.html>.

Customer Alliance. (n.d.). A Brief History of the Hotel Industry. [online] Available at: <https://www.customer-alliance.com/en/resources/article/hotel-industry/>.

Asksuite | Hotel Chatbot. (2021). The Evolution of Hotel Technologies: In-Room Phones to APIs. [online] Available at: <https://asksuite.com/blog/the-evolution-of-hotel-technologies/>.

Mailchimp. (2020). *Website Development: 8 Steps + Tips*. [online] Available at: <https://mailchimp.com/resources/guide-to-website-development/>.

Travelodge. (2023). Travelodge | Great value hotels throughout the UK & Ireland. [online] Available at: https://www.travelodge.co.uk/?gad_source=1&gclid=CjwKCAjw7SvBhB6EiwAwYdCAa1u534LwDppAQZ3H20p7RV iTTlMjMmUdkK8b8WcixrjQ6M0pQxhoCxYAAvD BwE&gclsrc=aw.ds [Accessed 19 Mar. 2024].

Marriott Bonvoy. (2024). Hotels & Resorts | Book your hotel directly with Marriott Bonvoy. [online] Available at: https://www.marriott.com/en-gb/default.mi?nst=paid&cid=PAI_GLB00050JC_GLE000BM3U_GLF000OP0J&ppc=ppc&pId=uktbppc&gad_source=1&gclid=CjwKCAjw7SvBhB6EiwAwYdCAa01PrTikHHQBhfSCYVZ0zD4IxBiJwQmvnopJGwjEaklidBAPKKALBoChrcQAvD BwE&gclsrc=aw.ds [Accessed 4 Mar. 2024].

Booking.com. (2019). Booking.com: The largest selection of hotels, homes, and holiday rentals. [online] Available at: <https://www.booking.com/index.en-gb.html>.

Figma. (2024). Figma. [online] Available at: <https://www.figma.com/file/h0r2TC0m3BgMMKAty9LtuA/Figma-basics?type=design&nodeid=4368-321189&mode=design> [Accessed 19 Mar. 2024].

Canva (2024). Canva. [online] Canva. Available at: <https://www.canva.com/>.

Rand, S. (2014). 24 Things to Consider When Designing and Developing a website. [online] social media Today. Available at: <https://www.socialmediatoday.com/content/24-thingsconsider-when-designing-and-developing-website>.

Copyscape.com. (2019). Copyscape Plagiarism Checker - Duplicate Content Detection Software. [online] Available at: <https://www.copyscape.com/>.

Morgan, N. (2023). JavaScript Crash Course. No Starch Press.

Powers, D. and Springerlink (Online Service (2010). PHP Solutions: Dynamic Web Design Made Easy. Berkeley, Ca: Apress.

www.adcisolutions.com. (2024). Web development methodologies and approaches | ADCI Solutions. [online] Available at: <https://www.adcisolutions.com/knowledge/webdevelopment-methodologies-and-approaches>.

Atlassian (2023). Waterfall Methodology for Project Management. [online] Atlassian. Available at: <https://www.atlassian.com/agile/project-management/waterfallmethodology>. [Accessed 9 Mar. 2024].

Microsoft (2016). Visual Studio Code. [online] Visual Studio Code. Available at: <https://code.visualstudio.com/docs/editor/whyvscode>.

Elegant Themes. (2022). A Quick Guide to phpMyAdmin (And How You Can Use It). [online] Available at: <https://www.elegantthemes.com/blog/resources/a-quick-guide-tophpmyadmin-and-how-you-can-use-it>.

edX. (2023). Learn XAMPP With Online Courses and Programs. [online] Available at: <https://www.edx.org/learn/xampp>.

Knight, A. (2022). What are HTML, CSS, & Javascript? [online] California Institute of Arts & Technology. Available at: <https://www.ciat.edu/blog/html-css-javascript/>.

GeeksforGeeks. (2020). What is PHP and Why we use it ? [online] Available at: <https://www.geeksforgeeks.org/what-is-php-and-why-we-use-it/>.

Naeem, T. (2019). Database Design - Overview, Importance, and Techniques. [online] Astera. Available at: <https://www.astera.com/type/blog/all-you-need-to-know-aboutdatabase-design/>.

GeeksforGeeks. (2022). How to Install Visual Studio Code on Windows? [online] Available at: <https://www.geeksforgeeks.org/how-to-install-visual-studio-code-on-windows/>.

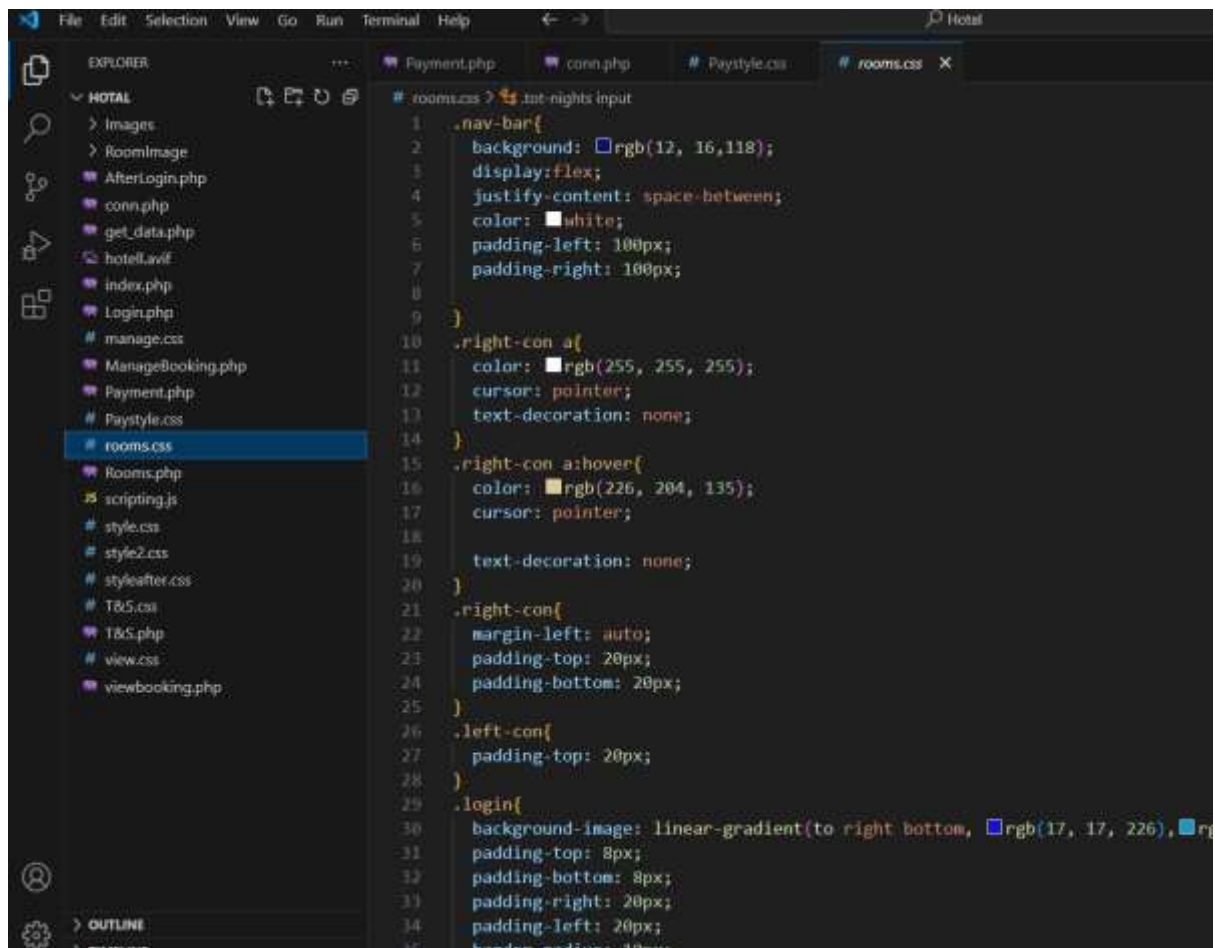
IONOS Digital Guide. (2023). How to install phpMyAdmin in a few steps. [online] Available at: <https://www.ionos.co.uk/digitalguide/websites/web-development/install-phpmyadmin/> [Accessed 26 Mar. 2024].

Redish, G. (2014). *Honing Your Usability Testing Skills*. [online] Available at: <https://redish.net/wp-content/uploads/Redish-Honing-Your-Usability-Testing-Skills.pdf> [Accessed 27 Mar. 2024].

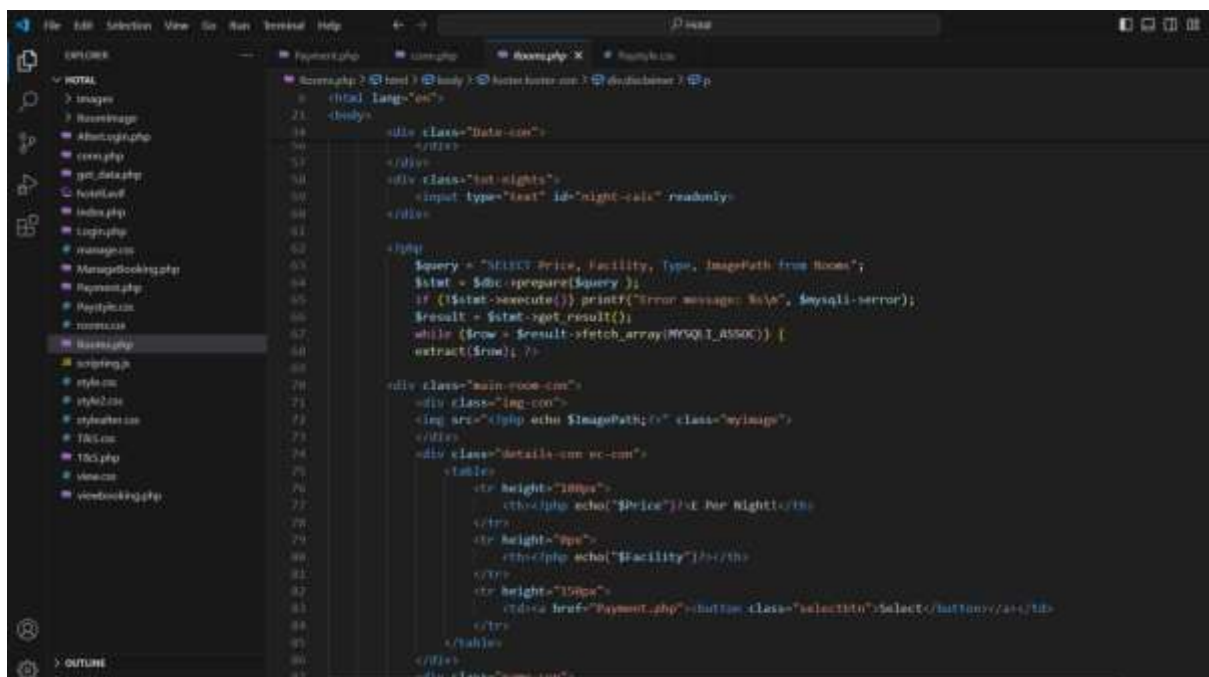
Reddish (2023). About Ginny | Ginny Redish. [online] Available at: <https://redish.net/about-ginny/> [Accessed 27 Mar. 2024].

9. Appendices

9.1 Screen shot of code:



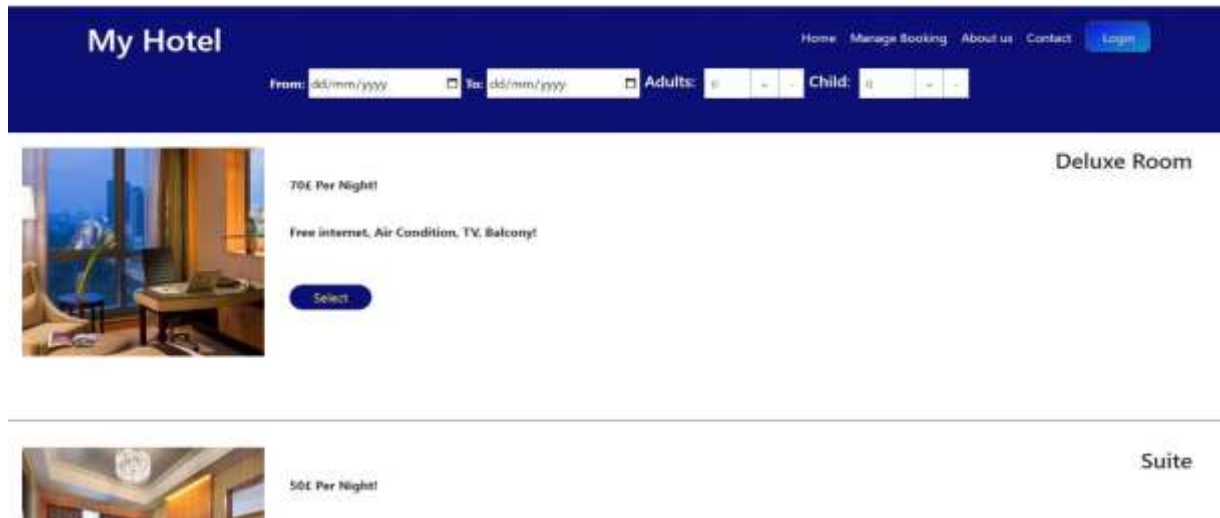
```
# rooms.css > int-nights input
1 .nav-bar{
2   background: #rgb(12, 16,118);
3   display: flex;
4   justify-content: space-between;
5   color: #white;
6   padding-left: 100px;
7   padding-right: 100px;
8 }
9
10 .right-con a{
11   color: #rgb(255, 255, 255);
12   cursor: pointer;
13   text-decoration: none;
14 }
15 .right-con a: hover{
16   color: #rgb(226, 204, 135);
17   cursor: pointer;
18
19   text-decoration: none;
20 }
21 .right-con{
22   margin-left: auto;
23   padding-top: 20px;
24   padding-bottom: 20px;
25 }
26 .left-con{
27   padding-top: 20px;
28 }
29 .login{
30   background-image: linear-gradient(to right bottom, #rgb(17, 17, 226), #rgb(17, 17, 226));
31   padding-top: 8px;
32   padding-bottom: 8px;
33   padding-right: 20px;
34   padding-left: 20px;
35   border-radius: 10px;
```

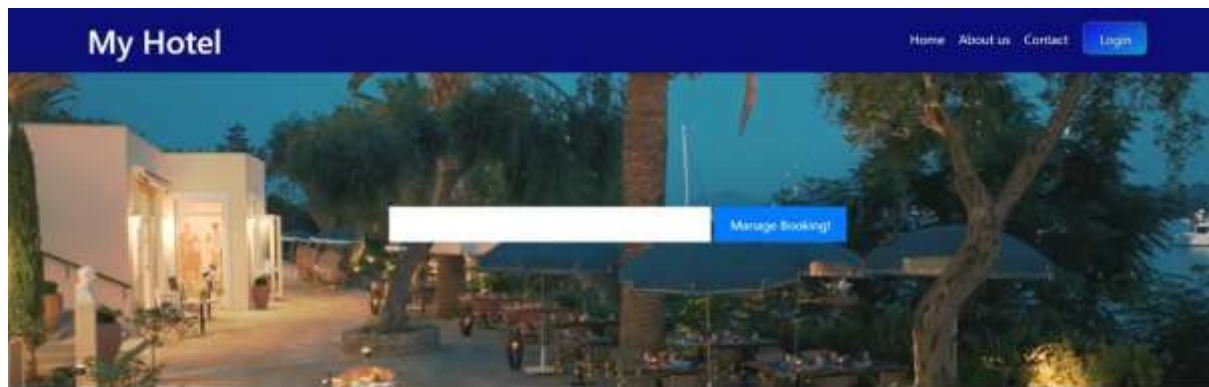



9.2 Screen shot of navigation bar:



9.3 Screen shot of the website:





Name: From: To:

Room type: Nights:

Per Night:

My Hotel
Home
Manage booking
About us
Contact
Login

Our Services

We elevate our clients above the crowd through integrity, intelligence and timely management. Our guests enjoy luxurious accommodations, exquisite dining options, and access to modern amenities including a fitness center, swimming pool, and spa. With a commitment to cleanliness and guest satisfaction, our dedicated staff is available around the clock to ensure a memorable and seamless stay.

Contact Us

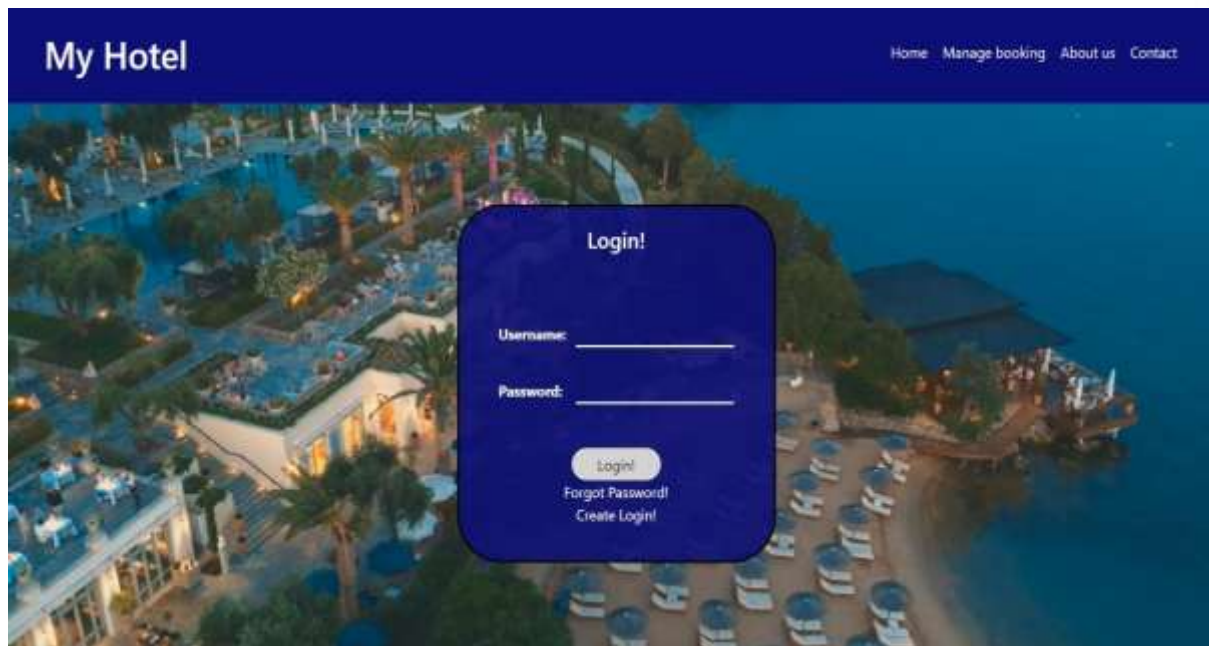
Phone no: 07456620385

Email: myhotel@mail.co.uk

Instagram: my_hotel_review

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Change Booking

Manage Booking

Name:

From:

dd/mm/yyyy

To:

dd/mm/yyyy

Room type:

Nights:

Per Night:

Email:

Contact:

Cancel Booking

Total Bookings

Booking ID	Customer Name	Room type	Per Night	Email	Contact	Nights	From	To
5595665	Qaisar Naem	Deluxe Room	70.00	qaidhq@gmail.com	7438802925	3	2024-03-07	2024-03-10
5595666	Hefla Joe	Single Room	35.00	heflajoe@gmail.com	7445233556	5	2024-04-10	2024-04-15